

In-Class Quiz: April 28

Points: 8/8

✓ **Correct** 1/1 Points

1. A processor has a 5-stage pipeline where each stage performs the same amount of work. If a non-pipelined version of the processor completes one instruction every T seconds, what is the exact execution time required to completely process 100 independent instructions in the pipelined processor? 

- ☐ $100T / 5$ seconds
- ☐ $100T$ seconds
- ☒ $T + 99(T/5)$ seconds
- ☐ $T/5 + 99T$ seconds

✓ **Correct** 1/1 Points

2. To ensure proper isolation and prevent interference between multiple instructions executing simultaneously, an "instruction packet" travels between stages. Based on the system design, which of the following is NOT explicitly defined as a component of this packet traveling across the pipeline latches? 

- ☐ Program counter
- ☒ Hardware memory sequence limits
- ☐ Control signals
- ☐ All intermediate results

✓ **Correct** 1/1 Points

3. How is the propagation of these instruction packets between the 5 stages strictly controlled in this datapath? 

- ☐ By 5 latches triggered by the positive edge of the clock.
- ☐ By 4 latches triggered by the positive edge of the clock.
- ☐ By 5 latches triggered by the negative edge of the clock.
- ☒ By 4 latches triggered by the negative edge of the clock.

✓ **Correct** 1/1 Points

4. The Operand Fetch (OF) stage is responsible for decoding instructions and preparing the necessary data for execution. Which specific bundle of signals is latched into the OF-EX register to be utilized by the Execute stage? 

- ☐ The program counter (pc), the instruction itself, the computed aluResult, and the control signal bundle.
- ☒ The program counter (pc), the computed branchTarget, the prepared ALU operands (A and B), the store operand (op2), the instruction itself, and the control signal bundle.
- ☐ The isBranchTaken flag, the calculated branchPC, ALU condition flags, and the prepared ALU operands (A and B).
- ☐ Memory control flags (isLd, isSt), the memory address register (mar), the memory data register (mdr), and the control signal bundle.

✓ **Correct** 1/1 Points

5. A pipelined processor heavily relies on control signals traveling alongside data. In which stage is the primary "Control unit" located that decodes the raw instruction to generate the control signal bundle? 

- ☐ Instruction Fetch (IF) Stage
- ☒ Operand Fetch (OF) Stage

- ☐ Execute (EX) Stage
- ☐ Register Write (RW) Stage

✓ **Correct** 1/1 Points

6. In the Execute (EX) stage, the processor must determine if a conditional branch should be taken. What specific outputs from the Arithmetic Logic Unit (ALU) are utilized by the Branch unit to make this logical determination? 

- ☐ The computed branchTarget
- ☐ The decoded aluSignals
- ☒ The generated flags
- ☐ The isUBranch control signal

✓ **Correct** 1/1 Points

7. When a Store instruction (isSt is active) reaches the Memory Access (MA) stage, it must provide an address and data to the Memory unit. Which values originating from the EX-MA pipeline register are mapped to the memory address register (mar) and memory data register (mdr)? 

- ☐ mar receives the program counter (pc); mdr receives the computed aluResult.
- ☒ mar receives the computed aluResult; mdr receives the designated store operand (op2).
- ☐ mar receives the first register operand (op1); mdr receives the designated store operand (op2).
- ☐ mar receives the computed branchTarget; mdr receives the loaded memory result (ldResult).

✓ **Correct** 1/1 Points

8. During a subroutine call (when isCall is active) in the Register Write (RW) stage, which specific value is selected by the multiplexer to be written into the return address register?



- ☐ The aluResult passed down from the EX stage.
- ☐ The ldResult fetched from the data memory in the MA stage.
- ☒ The instruction's pc plus 4, calculated by a dedicated adder located within the RW stage.
- ☐ The branchTarget generated dynamically in the OF stage.



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